Arista Access Node Failure SOP - HSR-ACC-27 - Optical Transport Failure

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1207

Description

The failure of the optical transport mechanism in the Arista access node HSR-ACC-27 at the HSR Layout Access site is attributed to the degradation of the SFP/XFP transceivers, fiber cuts, or a loss of receive/transmit power. This is due to the physical layer specifications of the Arista access node's 10Gigabit Ethernet ports, which rely on the SFP/XFP modules to establish and maintain optical connections with the adjacent nodes. The use of SFP/XFP modules with degraded performance or the occurrence of fiber cuts can lead to a loss of signal quality, resulting in a major alarm (ALARM_1207) due to the failure of the optical transport mechanism. The physical layer protocols, such as IEEE 802.3z, operate at the data link layer and rely on the optical transport mechanism for the transmission and reception of data. Therefore, any issues with the optical transport mechanism can have a cascading effect on the entire network, impacting the reliability and performance of the Arista access node.

Pre-requisite Checks

- Verify that the site location, HSR Layout Access, is accessible and secure for remote hands.
- Check the redundant link status to ensure that the link is active and available.
- Confirm that the Arista access node, HSR-ACC-27, is powered on and has a stable IP address.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts or damage	Medium
Loss of receive/transmit power	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface GigabitEthernet 1/1/1, Expected Output: GigabitEthernet1/1/1 is up, line protocol is up, Description: This command checks the status of the GigabitEthernet interface, verifying that it is up and the line protocol is up, which indicates that the interface is operational and can be used for diagnostics.

Command: show interface GigabitEthernet 1/1/1 transceiver, Expected Output: Transceiver is SFP-10G-SR, Description: This command displays the transceiver type and status for the specified interface, which can help identify any issues with the SFP/XFP module.
```

```
Command: show interface GigabitEthernet 1/1/1 counters, Expected Output: Input errors: 0, Output errors: 0, Description: This command displays the error counters for the specified interface, which can help identify any issues with the optical transport mechanism.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: replace interface GigabitEthernet 1/1/1 with SFP-10G-SR, Description: Replace the SFP/XFP transceiver with a new one to resolve the issue. This step requires physical access to the Arista access node and should be performed by a qualified engineer.

Step 2: Step: Verify the fiber connection, Command: show interface GigabitEthernet 1/1/1 transceiver, Description: Verify that the fiber connection is secure and not damaged, which can cause issues with the optical transport mechanism.

Step 3: Step: Reset the interface, Command: reset interface GigabitEthernet 1/1/1, Description: Reset the interface to its default settings and re-establish the connection to resolve any issues with the optical transport mechanism.

Post-Check Verification

```
Command: show interface GigabitEthernet 1/1/1, Expected Output: GigabitEthernet1/1/1 is up, line protocol is up, Description: Verify that the interface is up and the line protocol is up after resolving the issue.

Command: show interface GigabitEthernet 1/1/1 counters, Expected Output: Input errors: 0, Output errors: 0, Description: Verify that the error counters are zero after resolving the issue.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: replace interface GigabitEthernet 1/1/1 with original SFP-10G-SR, Description: Restore the original SFP/XFP transceiver to its previous state in case the replacement did not resolve the issue.

- Step: Reset the interface to its previous state, Command: reset interface GigabitEthernet 1/1/1 to previous state, Description: Reset the interface to its previous state in case the resetting of the interface did not resolve the issue.